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***B.Tech. Degree VII Semester Supplementary Examination
April 2018***

**ME 1702 VIBRATION AND NOISE CONTROL
(2012 Scheme)**

Time: 3 Hours

Maximum Marks: 100

**PART A
(Answer *ALL* questions)**

(8 × 5 = 40)

- I. (a) A 0.453 kg mass attached to a light spring elongates it by 7.87 mm. Determine natural frequency of the system.
- (b) A harmonic motion has an amplitude of 0.2 cm and a period of 0.15 s. Determine the maximum velocity and acceleration.
- (c) Explain the theory behind the working of seismometer and accelerometer.
- (d) Explain the method of converting a geared system into an equivalent three rotor torsional system for determining the natural frequencies.
- (e) Derive the equation for the natural frequencies of a uniform cord of length l fixed at the two ends. The cord is stretched to a tension T and its mass per unit length is ρ .
- (f) Compare Rayleigh's and Dunkerley's methods in determining the natural frequency of transverse vibrations of a shaft carrying two point loads.
- (g) Differentiate between loudness and frequency.
- (h) What is an acoustic spectra? Explain.

PART B

(4 × 15 = 60)

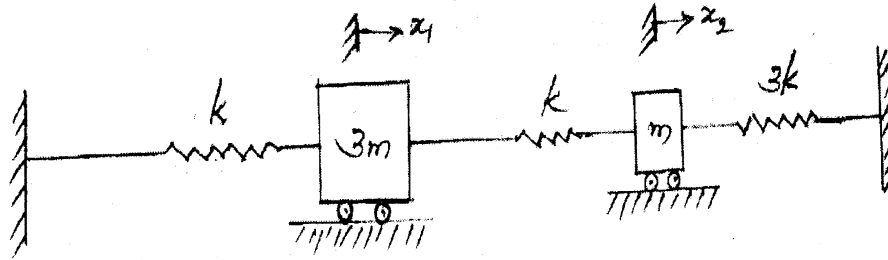
- II. (a) Determine the decay of amplitude for a vibrating system with Coulomb damping. (5)
- (b) An unknown mass m kg attached to the end of an unknown spring of stiffness k has a natural frequency of 94 cycles per minute. When a 0.453 kg mass is added to m , the natural frequency is lowered to 76.7 cycles per minute. Determine the unknown mass m (kg) and the spring stiffness k (N/m). (10)

OR

- III. (a) A mass of 0.907 kg is attached to the end of a spring with a stiffness of 7 N/cm. Determine the critical damping coefficient. (5)
- (b) A vibrating system consists of a mass of 4.534 kg, a spring of stiffness 35 N/cm and a dashpot with a damping coefficient of 0.1243 N/cm/s. Determine: (10)
 - (i) Damping factor.
 - (ii) The logarithmic decrement.
 - (iii) The ratio of any two consecutive amplitudes.

(P.T.O.)

- IV. (a) Setup the equations of motion of a double pendulum in terms of angle θ_1 and θ_2 , measured from the vertical. Mass $m_1 = m_2 = m$ and length $L_1 = L_2 = L$. (5)
- (b) Determine the natural frequencies and mode shapes of the system shown in figure. (10)



OR

- V. Two rotors A and B are attached to the ends of a shaft 600 mm long. The mass of rotor A is 400 kg and its radius of gyration is 400 mm. The corresponding values of rotor B are 500 kg and 500 mm respectively. The shaft is 80 mm diameter for the first 250 mm length, 120 mm for the next 150 mm length and 100 mm for the remaining length. The modulus of rigidity of the shaft material $G = 80 \text{ GN/m}^2$. Determine: (15)
- The position of node.
 - The frequency of torsional vibrations.

- VI. Derive the frequency equation for a uniform beam fixed at both ends and having transverse vibrations. (15)

OR

- VII. A hollow shaft 1.8 m long is supported in short flexible bearings at the ends. It carries two wheels each of 60 kg mass, one at the centre of the shaft and the other at 450 mm from the centre. The external and internal diameters of the shaft are 80 mm and 50 mm respectively. Determine the natural frequency of transverse vibration of the shaft. The density of the shaft material is 7500 kg/m^3 . Take $E = 210 \text{ GN/m}^2$. (15)

- VIII. (a) What is an octave band? Differentiate between full octave and one-third octave bands. (7)
- (b) Three machines operating individually make sound pressure levels of 90 dB, 88 dB and 97 dB respectively. Determine the overall sound pressure level when they operate simultaneously. (8)

OR

- IX. (a) Show that the sound intensity level decreases by 6 dB for spherical acoustic waves, as the distance from a point source is doubled. (7)
- (b) What are the recommended noise exposure limits? Plot the variation of noise exposure in decibel versus time. (8)
